SYNCHRONOUS COUNTER DESIGN

Step 1: Decide the no. & Type of Flip-Flops

Step 2: Excitation Table for the Flip-Flop

Step 3:- State Diagram & Ckt Exct table

Step 4:- Obtain Boolean expression

Step 5:- Draw Ckt Diagram.

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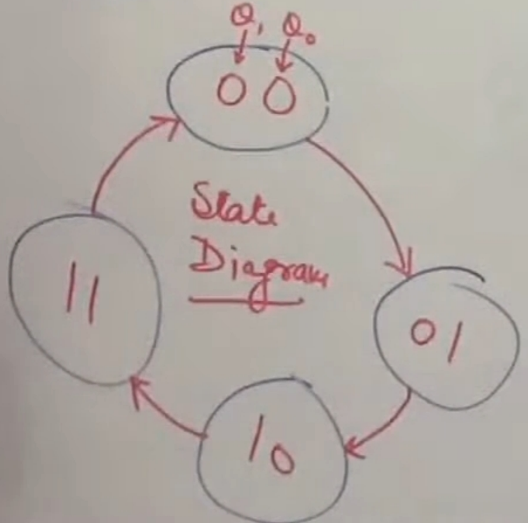
Step 3



Step 1: no. of FF = no. of bits

2 FF → JK FF

Step 2: Excitation Table

PS Q_n	NS Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

ex :- 2-Bit syn. Counter

Step 4 : Excitation Table & K-map for ckt :-

0/Ps 1 | X 0

PS		N.S		I/P			
Q_1	Q_0	Q_1	Q_0	J_1	K_1	J_0	K_0
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

J_1

$Q_0 \backslash Q_1$	0	1
0	0	X
1	1	X

$J_1 = Q_0$

J_0

$Q_0 \backslash Q_1$	0	1
0	1	1
1	X	X

$J_0 = 1$

$Q_0 \backslash Q_1$	0	1
0	X	0
1	X	1

$K_1 = Q_0$

$Q_0 \backslash Q_1$	0	1
0	X	X
1	1	1

$K_0 = 1$

Final Ckt diagram for 2 bit Sync. Counter